Number Theory — problem 6.100

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The product of two numbers of the form (a) $x^2 + 2y^2$ (b) $x^2 - 2y^2$ (c) $x^2 + dy^2$ (d) $x^2 - dy^2$ again has the same form (d is not a square).

Hint : $x^2 - dy^2 = (x + y\sqrt{d})(x - y\sqrt{d})$, $x^2 + 2y^2 = (x + iy\sqrt{d})(x - iy\sqrt{d})$.